



CLASH-OF-LEGES: A bilingual dataset for conflict detection and explanation in statutory law

Paolo Italiani ^{a,1}, Gianluca Moro ^{a,1,*}, Luca Ragazzi ^{a,1}

^a Department of Computer Science and Engineering - DISI, University of Bologna, Via dell'Università 50, Cesena, 47522, Italy

ARTICLE INFO

Keywords:

Legal AI
Legal conflict detection
Multilingual dataset
Constitutional law
Large language models

ABSTRACT

Legal conflicts between statutes or constitutional articles present a significant challenge in maintaining consistency and coherence within legal systems. Addressing these conflicts requires extensive human expertise, making the process labor-intensive and time-consuming. In this paper, we introduce CLASH-OF-LEGES, a novel multilingual dataset derived from rulings by the Constitutional Court of the Italian Republic, designed to aid the automation of conflict detection and explanation between legal articles. We identify three key tasks: Conflict Classification, which determines whether two legal articles are in conflict; Conflict Explanation Generation, which provides detailed explanations for identified conflicts; and Reference Retrieval, which sources relevant legal bases and precedents to substantiate interpretations. These tasks are intended to facilitate the development of AI models that can automatically identify and explain contradictions between legal provisions.²

1. Introduction

The integration of artificial intelligence (AI) into the legal domain is reshaping longstanding practices, from legal research and case analysis to drafting and decision-making. While traditionally resistant to automation, the legal sector is increasingly adopting AI tools that bring efficiency, consistency, and accuracy to complex tasks. Among AI's primary contributions to legal practice are automated document review and legal forecasting. By employing natural language processing (NLP), AI systems swiftly identify pertinent legal statutes and precedents, allowing lawyers to streamline case preparation (Rissland & Ashley, 1987). Additionally, predictive analytics trained on historical case data enable lawyers to anticipate case outcomes, helping them evaluate case strength and advise clients with greater confidence. For instance, models trained on past court decisions have shown promising results in predicting outcomes in areas such as bankruptcy, intellectual property, and contract disputes (Aletras et al., 2016). AI-driven applications now assist also with document summarization (Moro et al., 2023a; Moro & Ragazzi, 2022, 2023; Moro et al., 2023c; Ragazzi et al., 2024, 2025) and question answering (Italiani et al., 2025) for case law analysis, and contract management (Ashley, 2017) by parsing vast legal texts, revealing patterns, and providing rapid, precise insights that enhance productivity and reduce costs. These advances underscore AI's potential to perform

highly nuanced tasks in law that, until recently, were believed to require human expertise exclusively.

Despite these advancements, the process of drafting new laws remains an underexplored area of AI application, particularly in addressing legal conflicts. Crafting new statutes requires rigorous checks to prevent overlaps or contradictions with existing laws. Conflicting statutes or ambiguous regulations can destabilize legal interpretations, increase litigation, and place burdens on judicial systems tasked with resolving these issues. Traditionally, legislative drafters consult constitutional experts and subject the laws to thorough reviews to minimize conflicts. In some jurisdictions, like Italy, the Constitutional Court plays a crucial role in reviewing new laws to ensure they align with constitutional principles. Yet, even with such safeguards, legal conflicts frequently arise within complex legal systems, complicating judicial processes.

Once identified, these conflicts must be resolved by courts to clarify statutory hierarchies and maintain legal coherence. For instance, in the Italian legal system, the Constitutional Court often deals with conflicts between national laws and constitutional articles, issuing rulings to clarify the hierarchy and proper interpretation of competing legal provisions. However, manually detecting and resolving such conflicts is time-intensive and demands significant legal expertise. Legal professionals must sift through extensive legal texts, interpret overlapping provisions, and cross-reference constitutional frameworks, a

* Corresponding author.

E-mail addresses: paolo.italiani@unibo.it (P. Italiani), gianluca.moro@unibo.it (G. Moro), l.ragazzi@unibo.it (L. Ragazzi).

¹ These authors contributed equally to this work (co-first authorship).

² The code is available at <https://github.com/disi-unibo-nlp/clash-of-leges>

Table 1

Summary of the most relevant conflict detection benchmarks. Unlike prior datasets, CLASH-OF-LEGES is grounded in authentic legislative jurisprudence and supports multilingual, cross-document, and explanatory tasks for statutory conflict analysis.

Name	Annotation	Tasks ^a	Cross-Doc. ^b	Languages	# Confls. ^c	Type
CONTRACTNLI (Koreeda & Manning, 2021)	Manual	NLI	✗	English	841	Contracts
NCI-EN (Aires et al., 2019d)	Semi-synthetic	MLC	✗	English	228	Contracts
NEURALCONFLICT (Huang et al., 2024b)	Synthetic	BC	✗	English, Chinese	6230	Contracts, Policies
Aires et al. (Aires & Meneguzzi, 2021)	Semi-synthetic	BC	✗	English	104	Contracts
CLASH-OF-LEGES (Ours)	Manual	BC ^d , CE ^d , AR	✓	Italian, English	632	Legislations

^a BC = Binary Classification, MLC = Multi-Label Classification, NLI = Natural Language Inference, CE = Conflict Explanation, AR = Article Retrieval.

^b ✓ = the potentially conflicting texts belong to different documents; ✗ = otherwise.

^c Number of annotated conflicting pairs.

^d LLM evaluation.

process that becomes increasingly challenging as the body of law expands.

Existing works have evident limitations (see Table 1). As outlined by Schumann and Gómex (2024), only two studies proposed datasets containing more than 250 cases of conflict, and the vast majority of conflict cases (87 %) were (semi-)synthetically generated. This reliance on synthetic conflicts raises concerns about the generalizability of these datasets, as they may not accurately reflect the complexity and nuances of real-world conflicts. Furthermore, half of all studies demonstrated their approaches using only a few individual examples rather than evaluating them holistically with annotated test datasets. Such limited evaluations make it challenging to assess the robustness and applicability of these methods in practical scenarios. Finally, in terms of addressed tasks, current datasets primarily focus on classification, without supporting

downstream tasks like conflict explanation or the retrieval of relevant legal provisions—both essential for real-world legal applications.

Our work addresses these gaps by directly engaging with real-world legal rulings from the Constitutional Court of the Italian Republic (CCIR). In particular, we present CLASH-OF-LEGES, a novel bilingual dataset designed to assist AI models in detecting and explaining conflicts within statutory law (see Fig. 1). By focusing on rulings where constitutional conflicts are explicitly identified, our dataset provides a realistic and practical foundation for evaluating detection approaches. This emphasis on real-world data ensures that the models developed using our dataset are better equipped to handle the complexity of conflicts as they occur in practice, bridging the gap between theoretical frameworks and practical legal applications. Additionally, we also incorporate an automatically translated version of the dataset in English to enhance its accessibility and broaden its utility for international research. Based on this resource, we introduce three core tasks essential for addressing legal conflicts in drafting and statutory review:

- **Conflict Detection:** Identifying whether two legal provisions conflict.
- **Conflict Explanation Generation:** Providing automated explanations for detected conflicts.
- **Article Retrieval:** Sourcing relevant legal bases and precedents to substantiate interpretations.

We perform comprehensive quantitative analyses on CLASH-OF-LEGES, utilizing a broad array of evaluation metrics to pave the way for future research developments. In particular, we systematically evaluate the performance of state-of-the-art multilingual large language models (LLMs) under practical limitations, such as running on a single 24 GB GPU. Our study evaluates these models' adaptability across languages and tasks, revealing that significant improvements are needed for practical usability.

2. Related work

2.1. Datasets and jurisdictions

In recent years, a diverse set of legal NLP benchmarks has emerged, reflecting the structural, linguistic, and jurisdictional heterogeneity of legal systems. Within common law systems, widely adopted datasets include CaseHOLD (Zheng et al., 2021) for legal judgment prediction in the U.S. context and ECtHR (Chalkidis et al., 2021a) for classification and outcome prediction in rulings by the European Court of Human Rights. In addition, the COLIEE task, based on Canadian case law, supports legal question answering and textual entailment across multiple subtasks. Conversely, in civil law systems, datasets are more often centered on statutory interpretation and normative reasoning. Notable examples include LeCARD (Ma et al., 2021), a Chinese dataset for legal

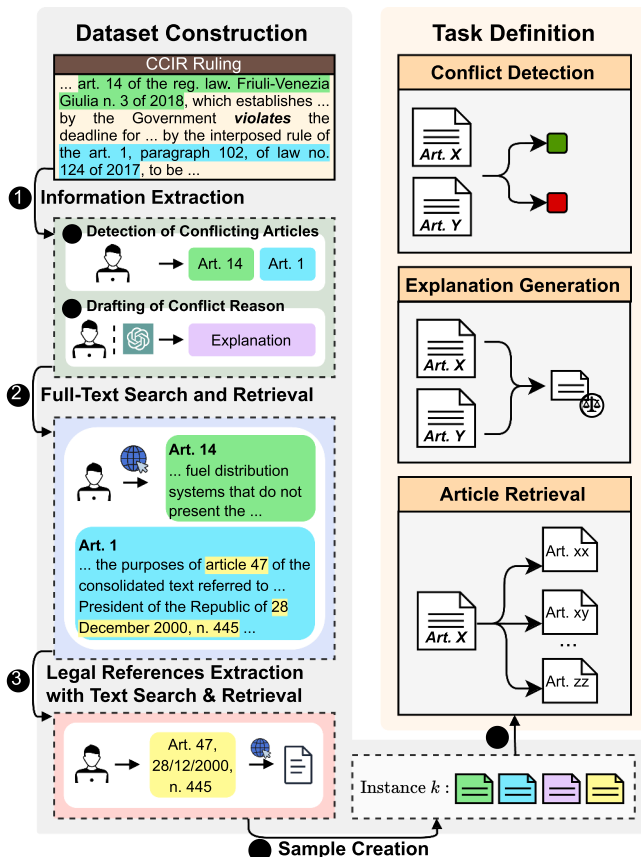


Fig. 1. Overview of CLASH-OF-LEGES.

reading comprehension, LEDGAR (Tuggenier et al., 2020), which provides clause-level annotations from commercial contracts under Swiss law, and LAWSUIT (Ragazzi et al., 2024), a summarization dataset derived from Italian Constitutional Court rulings.

2.2. Conflict detection datasets

Only a limited number of corpora provide structured annotations for legal conflicts. Most benchmarks, such as ContractNLI (Koreeda & Manning, 2021) and NeuralConflict (Huang et al., 2024b), focus on contract law and rely on synthetic or semi-synthetic conflict pairs, with an exclusive emphasis on English and common law contexts. Some efforts target normative or policy documents (Zadrozny et al., 2017), but they rarely include real-world judicial data, constitutional review, or multilingual content. Similarly, works like Aires et al. (2019a) concentrate on clause-level norm conflicts using semantic representations and entailment-based reasoning, assuming monolingual, decontextualized settings that reflect common law interpretive patterns. Exceptions include the alignment of EU directives with Member State laws (Araszkiewicz et al., 2021), or compliance between SOPs and medical guidelines (Zadrozny et al., 2017), yet these are often domain-specific, lack statutory focus, and are not framed within a structured legal hierarchy. In contrast, conflict detection in civil law systems remains largely underexplored. Existing approaches rarely engage with real-world statutory conflicts as they emerge in legislative or constitutional review. Our work addresses this gap by introducing a bilingual corpus grounded in Italian Constitutional Court rulings, supporting not only binary conflict detection as in the current benchmarks, but also explanation and reference retrieval, thereby enabling the development of interpretable, multilingual legal AI for civil law contexts.

2.3. Explanation generation in legal AI

Explanation generation is an increasingly prominent objective in legal AI, motivated by the need for transparency and trust. In tasks such as judicial outcome prediction (He et al., 2025), several efforts have focused on generating natural language rationales (Chalkidis et al., 2021b; He et al., 2023; Zhong et al., 2020) and legal entailment (Koreeda & Manning, 2021). These approaches typically rely on pre-trained language models fine-tuned to output structured or free-form explanations, sometimes augmented with attention or rationale extraction layers to enhance interpretability. Other works adopt rule-based or logic-driven templates, especially in highly structured domains like regulatory compliance (Rabelo et al., 2021).

Despite this growing interest, explanation generation in the context of statutory conflict detection remains underexplored. Existing datasets often lack explicit annotations of the legal reasoning behind conflicts, and few benchmarks support multilingual, cross-document, or constitutionally-grounded explanation tasks. Our work addresses this gap by explicitly framing conflict explanation as a supervised task, pairing each identified conflict with concise rationales extracted from real-world constitutional rulings. This enables the training and evaluation of models that not only detect contradictions, but also articulate the legal principles and statutory inconsistencies that underlie them.

2.4. NLP Methods for legal conflicts

The task of identifying conflicts within legal texts aligns closely with broader research in the domain of consistency checking and conflict detection across textual content. One prominent approach involves the use of formal logic systems to model and analyze regulatory texts (Chittimalli & Anand, 2016; Chittimalli et al., 2019; Mitra et al., 2018). By representing regulations through logical expressions, conflicts can be identified by detecting logical inconsistencies. Deontic logic, which deals with obligation, permission, and prohibition, has been particularly useful in this context. This method allows for the formalization of norms

and the automated detection of normative conflicts within and between regulations.

NLP techniques have also garnered significant attention, and one of the most prevalent approaches involves training classifiers in a supervised fashion. Convolutional Neural Networks have been widely used for this task, as seen in studies such as Aires et al. (2019c), Aires and Meneguzzi (2021, 2017). Other architectures, such as Long Short-Term Memory (LSTM) networks, have also been applied to norm conflict detection. For example, studies like Alshamsan and Chaudhry (2022), Huang et al. (2024b) utilized LSTMs for this purpose. Similarly, Support Vector Machines have been leveraged in various works, including Aires et al. (2019b,d), highlighting their effectiveness in classification tasks related to norm conflicts. Moreover, transformer-based models have gained attention in recent works, including Ichida and Meneguzzi (2021), van Leijenhof et al. (2023).

In order to train such supervised solutions, several methodologies have been explored to construct pairs of conflicting texts. One prevalent approach focuses on sentence pairing based on similarity metrics. Tools employing sentence embeddings or n-gram-based cosine similarity have been utilized in Hematitalam (2021), van Leijenhof et al. (2023), Sai et al. (2023). These methods involve linking sentences into candidate pairs once a similarity threshold is met. For example, in Sai et al. (2023), a two-step pairing process was applied to align sentences from hospital's standard operating procedures (SOPs) with recommendations from medical guidelines. Similarly, term-based or entity-based pairing approaches were employed in Aires et al. (2018, 2017), Araszkiewicz et al. (2023), where only provisions concerning the same parties were compared. Additionally, techniques like TextDiff functions, as used in Abualhaija et al. (2024), enabled pair creation by identifying differences or absence between provisions in separate lists. Semi-synthetic datasets have also been a significant focus in conflict detection research, with studies such as Aires et al. (2019b,d), Aires and Meneguzzi (2021), Huang et al. (2024b) that generated semi-synthetic evaluation datasets by inserting contradictions into conflict-free texts. This was achieved by manually or semi-automatically altering random normative sentences to create annotated sentence pairs for model training. Other studies, such as Alshamsan and Chaudhry (2022), Koreeda and Manning (2021), constructed a Cartesian product of all relevant text components and subsequently performed their analysis on the resulting pairs.

While the aforementioned studies primarily compare information within similar document types, some have extended this to cross-type document comparisons. Examples include the alignment of SOPs with medical guidelines (van Leijenhof et al., 2023), EU Directives with Member State laws (Araszkiewicz et al., 2021), and Data Protection Guidelines with the GDPR (Alshamsan & Chaudhry, 2022). These inter-document comparisons illustrate the potential for identifying conflicts across heterogeneous legal sources.

3. CLASH-OF-LEGES

The dataset is constructed from rulings issued by the Italian Constitutional Court, focusing on cases where the constitutional legitimacy of specific articles was evaluated (see Fig. 3 for the annual distribution of

Table 2

Text statistics for all samples generated using NLTK (Bird & Loper, 2004), reporting the average number of words (w) and sentences (s) across potentially conflicting article pairs (Art. Pair), conflict explanations (Explan.), and articles employed in the retrieval task (Retr.).

	Art. Pair		Explan.		Retr. Art.	
	w	s	w	s	w	s
Italian	325.68	8.73	48.10	1.53	163.25	4.15
English	338.00	9.26	48.54	1.54	173.35	4.63

EXAMPLE 1	
CONFL. PAIR	[ART] For the purposes referred to in article 42, paragraph 6, of regional law 19/2012, fuel distribution systems that do not present the adaptation program to the Municipality or of closing the plant within two years from the date of entry into force of this law. [CONFL. ART] At the same time as registration in the registry referred to in paragraph 100 of this article, the owners of fuel distribution systems must submit a declaration in lieu of the affidavit, made pursuant to and for the purposes of article 47 of the consolidated text referred to in the decree of the President of the Republic of 28 December 2000, n. 445, addressed to the Ministry of Economic Development, the competent region, the administration responsible for issuing the authorization or concession title and the office of the Customs and Monopolies Agency, certifying that the fuel distribution plant falls or does not fall, in relation only to aspects relating to road traffic safety, in one of the cases of incompatibility provided for by current regional provisions and better specified, for the sole purpose of submitting the declaration in lieu of the affidavit, in paragraphs 112 and 113 of this article, or who, despite falling within the cases of incompatibility, undertake to adapt them, to be completed within eighteen months from the date of entry into force of this law. Within fifteen days of the conclusion of the adaptation works, the holder of the authorization presents a declaration in lieu of the affidavit relating to the compatibility of the system referred to in this paragraph. The declaration referred to in the previous period may be accompanied by a formal exemption, arranged prior to the date of entry into force of this law by the competent administration on the basis of specific regional regulations. As an alternative to the aforementioned declaration, a sworn appraisal by a qualified technician can be provided.
EXPLAN.	The regional law introduces a deadline for the adaptation or closure of fuel distribution systems in contrast with the state deadline, violating the exclusive competence of the State in matters of road safety.
EXAMPLE 2	
CONFL. PAIR	[ART] Pending the approval of the resolutions referred to in this article, the issuing of new authorizations for wind farms in the regional territory is suspended. In order to avoid circumvention of the legislation for the protection of the environment, cultural heritage, health and public safety, without prejudice to the provisions of Part Five of Legislative Decree 3 April 2006, n. 152, and subsequent amendments, and, in particular, from articles 270, 273 and 282, as regards the identification of plants and the conveyance of emissions, the Regions and autonomous Provinces establish the cases in which the presentation of multiple projects for the construction of plants powered by renewable sources and located in the same area or in adjacent areas are to be evaluated in cumulative terms as part of the environmental impact assessment. [CONFL. ART] The authorization referred to in paragraph 3 is issued following a single procedure, in which all the interested Administrations participate, carried out in compliance with the principles of simplification and with the methods established by law 7 August 1990, n. 241, and subsequent amendments and additions. The issuing of the authorization constitutes the right to build and operate the plant in accordance with the approved project and must contain, in any case, the obligation to restore the state of the premises to pristine condition which is the responsibility of the operator following the decommissioning of the plant. The maximum term for the conclusion of the procedure referred to in this paragraph cannot in any case be more than one hundred and eighty days.
EXPLAN.	The rule censored by the Campania TAR produces a procrastination effect which contrasts with the fundamental principle expressed by the art. 12, paragraph 4, of Legislative Decree no. 387 of 2003 regarding "production, transport and national distribution of energy".

Fig. 2. Two examples of conflicting article pairs, along with their explanations. The portions of the articles declared unconstitutional are highlighted in yellow, indicating their conflict with corresponding text portions of an existing law, which are highlighted in red. Additionally, the key elements of the explanations are emphasized in blue.

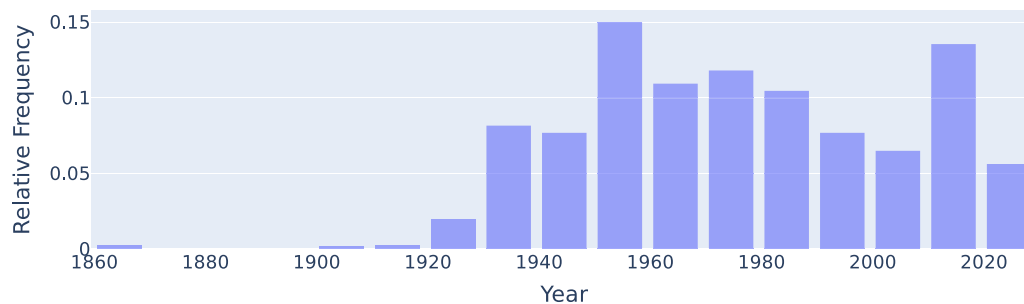


Fig. 3. Annual distribution of potentially unconstitutional articles.

instances). These rulings often involve a detailed analysis of legal conflicts, where one or more articles are found to contradict or challenge the principles set out in the Italian Constitution (see Fig. 4 for the instances' topic distribution). Table 2 reports the document statistics.

3.1. Dataset construction

Conflict Detection. The first step in constructing the dataset was identifying rulings where the CCIR assessed the constitutional legitimacy of one or more legal articles. In these cases, the Court determines whether the contested legal provisions align with the Italian Constitution, and if

not, the Court declares the provisions unconstitutional. We manually reviewed these rulings to extract instances where such constitutional conflicts were explicitly identified, forming the foundation of our dataset. All rulings were reviewed by a single legal expert and not by multiple annotators, since identifying conflicting provisions is largely objective. In fact, pairs of conflicting articles are immediately recognizable within each ruling due to the standardized language used to signal an antinomy. In addition, we also annotated cases where conflicts arose between two articles belonging to different laws. These conflicts often stem from situations where one legal provision challenges the validity or application of another, potentially creating indirect constitutional issues (see Fig. 2

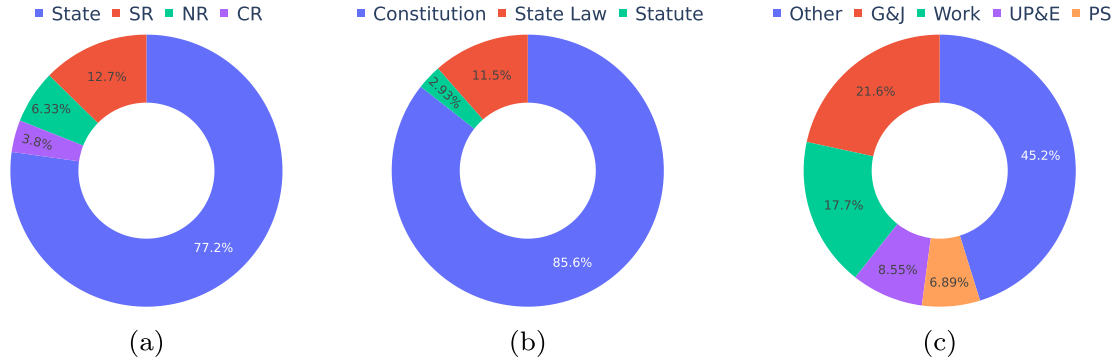


Fig. 4. Distribution of (a) potentially unconstitutional articles categorized by the authority that issued the law. The abbreviations represent regional classifications: SR (Southern Region), NR (Northern Region), and CR (Central Region); (b) legal instruments to which the articles potentially causing conflicts belong; (c) topics of potentially unconstitutional articles. G&J represents Governance and Justice, UP&E stands for Urban Planning and Environment, and PS refers to Public Services.

for two examples). By including both types of conflicts, our dataset captures a broader spectrum of legal discrepancies, encompassing not only direct constitutional disputes but also the interplay between different levels of statutory law. Fig. 4b illustrates the distribution of the various types of legal instruments involved in conflicts with the articles of law reviewed by the CCIR. Once rulings with constitutional conflicts were identified, the next step involved annotating the conflicts. In each ruling, legal articles identified as conflicting were meticulously annotated with their corresponding law number, year, article, and comma when referenced (see Fig. 1 step ①.(a) for the graphical breakthrough of our annotation approach). Then, to construct negative examples for the classification task—instances where two articles are determined not to be in conflict—we adopted a systematic approach instead of relying on simple random pairing. Specifically, we utilized CCIR rulings where constitutional illegitimacy claims were deemed unfounded. This method increases the task's complexity, presenting the model with pairs of articles that exhibit a plausible, albeit non-conflicting, relationship, rather than unrelated or at best loosely associated articles.

Conflict Explanation. The annotations focused also on the reasoning provided by the Court in its verdict. When available, this was extracted using regex rules; in cases where these rules proved insufficient, ChatGPT was employed to assist in the extraction operation (see Fig. 1 step ①.(b)). This process ensures that the dataset includes not only the conflicting articles but also provides valuable context about the nature of the conflict, which is critical for facilitating more accurate analysis of legal precedents and patterns across similar cases.

Reference Retrieval. The text of the articles may contain references to others, and in some cases, this contextual information is essential for accurate interpretation. The meaning of an article can easily be misinterpreted, as its implications often depend on its connections to related texts (Tiersma, 2000). Therefore, we automatically identify these references by leveraging the Akoma Ntoso standard,³ which provides a structured, domain-specific XML vocabulary for representing executive, legislative, and judicial documents in a standardized legal format. We define a retrieval task where textual references within the article text are removed, requiring the retriever model to accurately identify and return the correct referenced document from a database of 20,000 Italian legal articles sourced from Normattiva,⁴ a comprehensive repository of Italian legal texts (see Fig. 1 step ②).

³ <https://unsceeb-hlcm.github.io/>

⁴ The texts were freely obtained from <https://www.normattiva.it/> and are not considered authentic.

Table 3

The number of instances for each task organized into three splits. For the reference retrieval task, only a test set is provided, as no training is required for the experiments.

Task	Train	Validation	Test
Conflict Detection	1010	126	127
Conflict Explanation	201	25	26
Reference Retrieval	-	-	275

To complement the annotations with the full text of the conflicting articles and references, we collected the corresponding legal provisions from authoritative legal databases. For national laws, the articles were scraped from Normattiva. In cases where regional laws were involved, the relevant articles were retrieved from the regional legal websites, ensuring that the dataset includes a wide range of legal texts across different legislative levels (see Fig. 1 step ②). This multi-source approach ensures that the dataset is both broad and representative, covering a variety of conflicts between national and regional legal frameworks and the Constitution.

Table 3 presents the number of instances for each task, organized into training, validation, and test sets.

3.2. Language translation

We perform automatic translation of articles, references, and explanations from Italian to English by evaluating various machine translation models and services: GPT-4o, GOOGLE TRANSLATE, and LLAMA-3-8B. To assess translation quality, we employ two quality control methods and measure the effectiveness of the translations using a sample of 102 instances, evenly distributed across the training, validation, and test sets. In line with recent studies that circumvent human analysis due to its high cost (Feng et al., 2022; Scao et al., 2022), particularly when translating lengthy legal texts, we adopt a standard approach to assess translation effectiveness through Round-Trip Translation (or Back Translation). Technically, it involves translating a text into a different language and re-translating it to the original one. We implement the same approach as Feng et al. (2022) and Zhu et al. (2019) and employ two evaluation criteria. First, to assess syntactic accuracy, we compute ROUGE scores (Lin, 2004) by comparing the original text with its back-translated counterpart. Second, to evaluate beyond syntactic aspects, we measure sentence-level semantic similarity using an Italian cross-encoder.⁵

⁵ <https://huggingface.co/nickprock/cross-encoder-italian-bert-stsb>

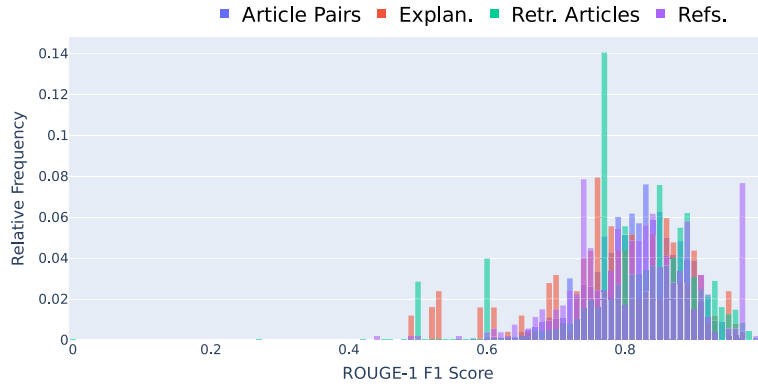


Fig. 5. GOOGLE TRANSLATE's back-translation performance evaluation on the different texts that compose the dataset.

Table 4

Average ROUGE and semantic similarity (SS) % scores on back-translation (full-text rulings) across all languages. The best results are in bold.

System	ROUGE-1	ROUGE-2	ROUGE-L	SS
GPT-4O	76.83	58.30	75.91	86.90
LLAMA-3-8B	66.80	44.81	66.36	83.01
GOOGLE TR.	82.68	68.62	83.18	88.17

The results depicted in Table 4 indicate that GOOGLE TRANSLATE offers the best overall performance in both syntactic and semantic dimensions, surpassing GPT-4O and LLAMA-3-8B. While GPT-4O provides strong results, particularly in terms of semantic similarity, LLAMA-3-8B lags behind in both ROUGE and semantic similarity scores. GOOGLE TRANSLATE is grounded in linguistic rules, making its translation more reliable. GPT-4O, on the other hand, while capable of producing fluent text, might prioritize coherence and context over preserving the original sentence structure, which could lead to discrepancies when the text is translated. For these reasons, GOOGLE TRANSLATE has been also the preferred choice in previous studies, which have successfully utilized it for multilingual content generation (e.g., BLOOMZ, T0) (Muennighoff et al., 2022) as well as for legal translations (Alkathery, 2023; Galassi et al., 2020; Ubayawardena & Hansani, 2023; WIESMANN, 2019). Fig. 5 presents the distribution of ROUGE-1 back-translation scores for GOOGLE TRANSLATE across the translated elements of the dataset, such as conflict pairs, explanations, referenced articles, and the articles used in the retrieval task.

To validate the reliability of the automatically translated dataset, we also conducted a human evaluation on a subset of instances. Specifically, we selected 127 translated article pairs from the test set of the conflict detection task and 26 translated explanations from the test set of the conflict explanation task. Each translation was independently assessed by two bilingual (Italian-English) annotators with legal expertise. Annotators rated each translation on a scale from 1 (poor) to 5 (excellent) according to the three dimensions defined in Table 5. The resulting inter-annotator agreement, measured by Cohens kappa, was 0.38. Table 6 reports the average scores across both annotators for the entire evaluated subset. Overall, the automatic translations achieved satisfactory performance across all three dimensions, with an average score across categories of 3.69 and 3.79 for article pairs and explanation, respectively. Importantly, the semantic-fidelity scores confirm that the original meaning is preserved.

Table 5

Evaluation dimensions for translated legal articles.

Dimension	Description
<i>Semantic Fidelity</i>	Does the translation accurately preserve the legal intent, obligations, rights and conditions of the original text?
<i>Legal Precision</i>	Are legal terms accurately translated and are article references and legal constructs well preserved?
<i>Clarity & Fluency</i>	Does the translation reflect the formal register and structured style typical of legal texts while remaining syntactically clear, natural, and easy to read in the target language?

Table 6

Human evaluation results on the article pairs and explanations belonging to the test set of the conflict classification and explanation tasks, respectively.

Data	Semantic Fidelity	Legal Precision	Clarity & Fluency	Avg.
Article Pairs	4.24	3.49	3.35	3.69
Explanations	4.31	3.63	3.44	3.79

4. Experiments

4.1. Experimental setup

Hardware Environment. All runs are tracked with Weights & Biases⁶ and executed on a workstation with a single Nvidia GeForce RTX3090 GPU of 24 GB of dedicated memory, 64 GB of VRAM, and an Intel® Core™i9-10900X1080 CPU @ 3.70GHz. The operating system is Ubuntu 20.04.3 LTS. To enhance consistency and portability, our development environment is built on top of a docker container with a NVIDIA image.⁷ All open-source models were fine-tuned on the training set of our dataset for 3 epochs using the Unsloth library (Daniel Han & team, 2023) to reduce VRAM consumption and enable faster fine-tuning, with the AdamW optimizer (learning rate 5e-6), a linear scheduler, and a fixed random seed of 42 for reproducibility; no preprocessing was performed.

Evaluation. We quantify classification effectiveness with recall, precision, F1-score, and accuracy. We evaluate conflict explanation

⁶ <https://wandb.ai>

⁷ <https://nvidia/cuda:12.3.2-devel-ubuntu22.04>

Table 7

Models source. Repository identifiers and access URLs for all models used in our experiments, covering both API-based and open-source releases.

Model	Snapshot / Repository	URL
<i>API Models</i>		
DEEPSEEK-R1	deepseek-reasoner	https://api-docs.deepseek.com/guides/reasoning_model
GPT-4o	gpt-4o-2024-11-20	https://platform.openai.com/docs/models/gpt-4o
<i>Hugging Face Models</i>		
GEMMA-2-9B	unsloth/gemma-2-9b-bnb-4bit	https://huggingface.co/unsloth/gemma-2-9b-bnb-4bit
MISTRAL-7B	unsloth/mistral-7b-v0.3-bnb-4bit	https://huggingface.co/unsloth/mistral-7b-v0.3-bnb-4bit
LLAMA-3-8B	unsloth/Meta-Llama-3.1-8B	https://huggingface.co/unsloth/Meta-Llama-3.1-8B
SAULLM-7B	Equall/Saul-7B-Base	https://huggingface.co/Equall/Saul-7B-Base

generation across several key dimensions. First, we report F1 scores for syntactic metrics, including ROUGE- $\{1,2,L\}$ (Lin, 2004), and the variance-aware aggregated score $\mathcal{R}$ (Moro et al., 2023b). Second, for semantic coverage, we employ the model-based metric BARTScore (Yuan et al., 2021), reporting precision, recall, and F1 values to capture nuanced performance. Finally, we report the Rank@ k metric, with $k \in \{1, 5, 10\}$ for the retrieval task. Technically, given an input article, we assess how frequently the corresponding references are correctly found within the top- k items retrieved.

Baselines. For the conflict classification and explanation tasks we consider LLAMA-3-8B,⁸ which represents a significant improvement over its predecessor, LLAMA-2, setting a new standard for LLMs of this scale through enhanced architecture and training techniques. GEMMA-2-9B (Team et al., 2024) delivers one of the best-in-class performances for its size, competing effectively with models 2-3 $\times$ larger, while remaining open and accessible to the community. MISTRAL-7B (Jiang et al., 2023) comprises grouped-query attention (Ainslie et al., 2023) and sliding window attention (Beltagy et al., 2020) to achieve a harmonious equilibrium between the pursuit of high performance and efficiency. SAULLM-7B (Colombo et al., 2024) builds upon the MISTRAL-7B architecture and is trained on a specialized English legal corpus comprising over 30 billion tokens. This tailored design achieves state-of-the-art performance in legal text comprehension and generation tasks. Additionally, we consider recent closed-source LLMs. Specifically, we employ GPT-4o (Hurst et al., 2024), which offers substantial improvements on text in non-English languages, while also delivering significantly faster inference and a 50% reduction in API cost. DEEPSEEK-R1 (DeepSeek-AI, 2025) is a Mixture-of-Experts language model optimized for reasoning tasks, combining supervised fine-tuning with large-scale reinforcement learning to achieve GPT-4-level performance with improved efficiency. The prompts that proved most effective for the conflict classification and explanation tasks are listed in Table 8. We deliberately omitted few-shot prompting because including even a minimal set of positive and negative examples would have made the input excessively long. Moreover, the diversity of conflict cases in our dataset is so varied that a single illustrative example would be insufficient to guide the model toward robust generalization. Table 7 provides the sources of all models used.

For the article retrieval task, we evaluate BM25OKAPI (Robertson et al., 1995), a probabilistic model that estimates document relevance with respect to a query as follows:

$$\text{BM25} = \sum_{i \in q} \log \left(\frac{n - f_i + 0.5}{f_i + 0.5} \right) \cdot \text{TF}_{\text{BM25}} \quad (1)$$

$$\text{TF} = \frac{f_{i,d} \cdot (k_1 + 1)}{f_{i,d} + k_1 \cdot ((1 - b) + (b \cdot \ell_d / \ell_{\text{avg}}))} \quad (2)$$

⁸ <https://ai.meta.com/blog/meta-llama-3/>

Table 8

Prompts employed for evaluating closed-source LLMs.

Task	System Prompt
Classification	You are an expert judge of the Italian Constitutional Court, charged with determining whether two legal articles conflict. You will receive input in this exact format: <full text of in <i>language</i> > [CONFL_ART] <full text of in <i>language</i> >. If the two articles conflict, respond exactly with YES. If they do not conflict, respond exactly with NO. Do not include any additional text, commentary, or explanation.
Explanation	You are an expert judge of the Italian Constitutional Court, charged with writing an explanation of why two articles conflict. You will receive input in this exact format: <full text of in <i>language</i> > [CONFL_ART] <full text of in <i>language</i> >. Write an explanation in <i>language</i> of maximum two sentences.

where n is the number of documents in the corpus, f_i is the number of documents containing the query term t , $f_{i,d}$ is the number of occurrences of t in the document d , ℓ_d is the number of terms in d , and ℓ_{avg} is the average of ℓ_d over the corpus. We use the standard parameters: $k_1 = 1.2$ and $b = 0.75$. predict relevance between queries and passages. Furthermore, we benchmark COLBERT-XM (Louis, Saxena), which effectively predicts relevance between queries and passages using only a limited set of English examples, leveraging the late interaction approach introduced in ColBERT (Khattab & Zaharia, 2020). Additionally, the model achieves effective zero-shot transfer across a diverse array of languages without requiring any further training. We also evaluate the *bi-encoder* (BiE) architecture (Reimers & Gurevych, 2019a), which employs a siamese BERT network to generate semantically meaningful sentence embeddings. In this framework, the query q and document d_i are independently processed through an encoder-only transformer, producing two distinct sentence embeddings that are subsequently compared using cosine similarity. To improve computational efficiency, we utilize FAISS (Johnson et al., 2019) for indexing, addressing the challenge of efficiently comparing high-dimensional vectors. As bi-encoder multilingual models, we operate with MPNET (Song et al., 2020), DISTILUSE (Reimers & Gurevych, 2019b), and MINILM (Wang et al., 2020). Finally, we explore a hybrid solution, denoted as HYBRID that fuses BM25OKAPI and MPNET scoring.

4.2. Results

Conflict Detection. The results presented in Table 9 show how, in the Italian setting, LLAMA-3-8B achieves the highest F1 score of 73.24, which suggests that it has a more effective understanding of the nuances and complexities inherent in Italian legal language. In the English dataset, LLAMA-3-8B continues to perform robustly, however

Table 9

Performance of baseline models on the conflict detection task. Bold denotes the best results.

Model	P	R	F1
IT			
🔥 GEMMA-2-9B	85.71	52.50	65.12
🔥 MISTRAL-7B	76.47	32.50	45.61
🔥 LLAMA-3-8B	83.87	65.00	73.24
🌟 GPT-4o	89.19	41.20	56.41
🌟 DEEPSEEK-R1	93.75	37.50	53.57
EN			
🔥 SAULLM-7B	84.21	40.00	54.24
🔥 GEMMA-2-9B	85.51	73.75	79.19
🔥 MISTRAL-7B	90.24	46.25	61.16
🔥 LLAMA-3-8B	87.50	70.00	77.78
🌟 GPT-4o	86.67	32.50	47.27
🌟 DEEPSEEK-R1	86.96	50.00	63.49

🔥 = fine-tuned; 🌟 = frozen

Table 10

Performance of baseline models on the article retrieval task. Bold denotes the best results.

Model	R@1	R@5	R@10
IT			
BM25OKAPI	8.99	19.33	22.06
DISTILUSE	5.69	13.30	16.71
MINILM	6.93	11.94	15.63
MPNET	5.69	16.63	22.01
COLBERT-XM	2.98	13.41	14.46
HYBRID	9.59	20.96	25.91
EN			
BM25OKAPI	6.72	18.14	20.79
DISTILUSE	6.57	14.57	17.89
MINILM	6.92	16.82	20.52
MPNET	6.56	18.32	21.73
COLBERT-XM	3.46	10.76	13.77
HYBRID	6.93	22.26	27.05

Table 11

Performance of baseline models on the conflict explanation generation task. Bold denotes the best results.

Model	R-1	R-2	R-L	R	BaS-R	BaS-P	BaS-F1
IT							
🔥 GEMMA-2-9B	20.21	4.95	22.30	15.73	-5.113	-5.476	-5.294
🔥 MISTRAL-7B	14.59	4.32	16.39	11.73	-6.744	-5.018	-5.881
🔥 LLAMA-3-8B	17.52	3.39	19.39	13.37	-5.646	-4.909	-5.278
🌟 GPT-4o	21.45	5.47	22.68	16.43	-4.379	-5.314	-4.846
🌟 DEEPSEEK-R1	20.30	4.41	21.10	15.18	-4.525	-5.236	-4.881
EN							
🔥 SAULLM-7B	22.01	3.57	22.56	15.92	-4.714	-8.139	-6.427
🔥 GEMMA-2-9B	32.80	8.75	28.71	23.16	-5.412	-4.871	-5.142
🔥 MISTRAL-7B	31.95	11.35	29.86	24.18	-5.909	-5.731	-5.820
🔥 LLAMA-3-8B	31.99	8.57	30.06	23.28	-5.459	-4.800	-5.129
🌟 GPT-4o	25.62	4.40	22.70	17.42	-6.629	-6.298	-6.463
🌟 DEEPSEEK-R1	22.57	3.35	21.20	15.59	-5.633	-5.996	-5.815

🔥 = fine-tuned; 🌟 = frozen

GEMMA-2-9B slightly surpasses it. Furthermore, a clear trend emerges, all source models, except for GPT-4o, perform better on the English dataset than on the Italian one, as shown in Fig. 8. For instance, GEMMA-2-9B's F1 score increases from 65.12 in Italian to 79.19 in English, and LLAMA-3-8B shows a similar improvement from 73.24% to 77.78%. This enhancement underscores the challenges posed by less-resourced languages like Italian in NLP tasks, particularly in specialized domains such as the legal one. The models generally exhibit higher Precision than Recall across both languages, implying they are more precise in

Table 12

LLM factuality evaluation prompt.

System Prompt
You are a legal expert evaluating the <i>factual accuracy</i> of a model's explanation of a conflict between two statutory articles in {language}.
Given:
•Article A: {article_a} Article B: {article_b}
•Model Explanation: {model_expl} Reference Explanation: {ref_expl}
Rate how factually correct the Model Explanation is compared to the Articles and the Reference Explanation, ignoring style or fluency.
Scoring: 5 Fully factual and legally accurate; 4 Mostly factual, minor omissions; 3 Partially correct, some inaccuracies; 2 Major factual or legal errors; 1 Mostly incorrect or irrelevant; 0 Completely wrong or hallucinated.
JSON only: {"score": <0-5>}

their conflict predictions but may not identify all instances of conflicting articles. This pattern is critical in legal contexts, where missing a conflict can have serious consequences. LLAMA-3-8B's and GEMMA-2-9B's relatively higher Recall suggests they are more reliable in capturing potential conflicts, which is advantageous for comprehensive legal analysis. Surprisingly, SAULLM-7B achieves the lowest F1 scores, indicating that its advanced understanding of legal language does not contribute in this task. Without task-specific adaptation, closed-source LLMs generate a high number of false negatives and fail to capture key distinctions that our benchmark emphasizes. In contrast, open-source LLMs that we fine-tune directly on our training set internalize the subtle patterns and edge cases that are encoded. With a more fine-grained analysis, Fig. 6 illustrates that the models generally perform better on the G&J and Work topics. Regarding legal instruments, the models demonstrate stronger performance in cases where the conflict is caused by state laws. This preference likely stems from the fact that identifying conflicts with constitutional articles demands a higher level of reasoning, as the cause of the conflict is often not explicitly stated in the text of the potentially conflicting pair. Furthermore, the improved performance in cases where the potentially unconstitutional article is issued by a regional authority may be attributed to the frequent involvement of state laws in such cases. Specifically, in 53.85% of cases where the issuing authority is a region, the potential conflict is caused by a state law.

Conflict Explanation. As showcased in Table 11, in both the Italian and English languages, GEMMA-2-9B consistently ranks among the top performers in R-1, R-2, and R-L scores. Notably, in the Italian task, GEMMA-2-9B achieved an R-1 score of 20.21, outperforming MISTRAL-7B, LLAMA-3-8B and SAULLM-7B. In the English task, the difference in ROUGE scores between the open-source models narrows, suggesting that MISTRAL-7B and LLAMA-3-8B may handle English explanations nearly as well as GEMMA-2-9B, particularly in R-L scores, which measure longer phrase matching. Across all models, English settings outperform Italian ones consistently in both ROUGE and BartScore metrics. For instance, GEMMA-2-9B's English R-1 score is 32.80, significantly higher than its Italian R-1 score of 20.21. This pattern confirms that the evaluated models, even if multilingual, are better optimized for English language tasks than Italian ones. Again, in line with the results observed in the classification task, we find a consistent performance gap between closed- and open-source models; however, on the Italian subset they converge to similar scores, with GPT-4o achieving the highest performance. The generally low ROUGE and BartScore metrics imply that generating coherent and accurate explanations about legal conflicts is complex and possibly still an open challenge for these models, especially in multilingual contexts. For a legal system, where detail accuracy is essential to explain conflicts between articles of law, these findings underscore the importance of continued fine-tuning, particularly for non-English language support. The differences in model performance across languages also highlight the need to assess model capabilities within the specific linguistic nuances and complexities of legal language, suggesting potential improvements in dataset balancing and model adjustments. To

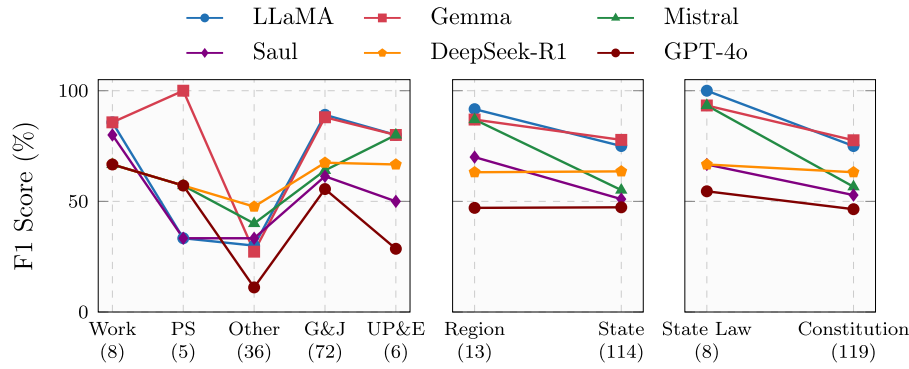


Fig. 6. Conflict detection F1 scores categorized by **left** topics of potentially unconstitutional articles. UP&E represents Urban Planning and Environment, G&J denotes Governance and Justice and PS refers to Public Services; **center** the legal instruments to which the articles potentially causing conflicts belong; **right** the authority that issued the potentially unconstitutional article. The number displayed beneath each label (n) indicates the count of instances associated with the corresponding label.

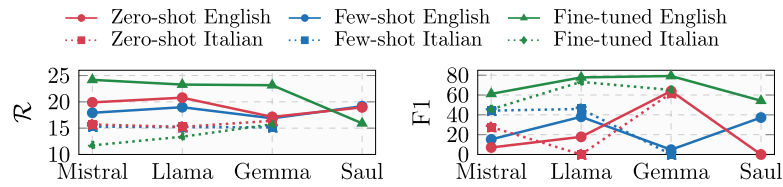


Fig. 7. Performance comparison across models in fine-tuning, zero-shot and few-shot settings. **Left** R score on explanation; **Right** F1 score on classification.



Fig. 8. Average metric scores by language for the (a) classification, (b) retrieval, and (c) explanation tasks. For the explanation task, only ROUGE-1/2/L scores are considered.

better understand the underlying causes of model failures, we conducted a detailed error analysis. A legal expert examined 10 randomly sampled examples for each of the three best-performing LLMs. Table 13 summarizes the most frequent types of errors identified in this evaluation. While the models generally succeed in identifying the broad source of the legal conflict, they exhibit critical flaws that undermine the reliability of their explanations. A recurring issue is the lack of *terminological precision*, which is especially problematic in the legal domain where accurate, context-specific language is essential and requires domain-specific knowledge. Another frequent error involves the application of a *wrong legal principle*. This misapplication reveals gaps in the models' understanding of foundational legal doctrines. We also observe instances of *missing conditions*, where essential legal qualifiers are omitted, resulting in incomplete or misleading reasoning. Finally, models exhibit errors related to the *subject matter*, either by referring to an entirely incorrect domain or by partially narrowing the scope of the legal issue at hand.

Zero and Few-Shot Analysis. To demonstrate the importance of task-specific adaptation, we compare the performance of open-source models in zero-shot and few-shot settings. The zero-shot experiments were conducted by prompting the models with the same instructions used for the closed-source LLMs, as shown in Table 8. In the few-shot setting, two examples with their corresponding labels were added to the prompt. Fig. 7 illustrates that fine-tuning consistently outperforms both zero- and few-shot settings across tasks and languages, with the only exception being the conflict explanation task on the Italian split, where differences are minor and overall performance remains poor. This finding further highlights the originality of our dataset. The knowledge acquired during pre-training alone is insufficient to achieve satisfactory performance, clearly indicating that our data lies outside the domain of the pre-training corpora. Moreover, the zero-shot setting consistently outperforms the few-shot setting, likely because two examples are insufficient to capture the overall data distribution and instead introduce additional noise into the prompt.

Table 13

Error analysis of the conflict explanation task, with highlighted text indicating the source of each identified error.

Model	Error Type	Prediction	Target	Details
GEMMA-2-9B	Wrong Legal Principle	that the contested provision provides for a discount on the fees for maritime state concessions, which are not owed by the subjects who have already regularized their situation, creating a disparity compared to those who have regularized the concession contract in another way, violating the principle of reasonableness and proportionality.	The contested provision provides for the facilitated settlement of pending proceedings only in favor of the State, excluding public bodies holding fees for the use of state-owned maritime assets, violating the principle of equality.	Prediction chose “reasonableness and proportionality” rather than the correct “equality” principle.
GEMMA-2-9B	Terminology Precision	which the fee is calculated on the basis of the efficient power rather than the nominal power of the plants, resulting in a more accurate and fair adjustment of the costs of the service, which are borne according to actual use rather than according to a standard parameter, which could also be different from the effective use.	the regional legislator - in the exercise of its competence, attributable to the concurrent power in the matter of “production, transport and distribution of energy” (art. 117, third paragraph, Cost), to determine the unit cost of the hydroelectric fee of users with nominal power exceeding 220 kW - has adopted a parameter, for the calculation of the overall fee due from the concessionaires, different from that of the “average nominal power”.	Prediction uses “efficient power” vs “nominal power,” but does not reference “average nominal power”, nor the threshold (220 kW).
LLAMA-3-8B	Terminology Precision	does not provide that the voluntary fish security guard qualification is not granted to those who have been convicted of a crime that entails the application of a prison sentence, even if suspended.	...which provides that the issuance and renewal of the qualification of voluntary fish protection guard is not granted to people sentenced to a prison sentence of no less than three years, when the sentence has become definitive or when it has been conditionally pardoned.	Unlike the target, which requires a minimum three-year sentence, the prediction imposes no minimum sentence length.
LLAMA-3-8B	Missing Condition	conflicts with the fundamental principles of the art. 1 of the Constitution, since it does not exclude from the possibility of benefiting from the procedure of collective land alienation, as a general rule, the owners of the land who are not in possession of it.	regulating the alienation of collectively owned land for civic use (so-called de-statementisation, or declassification of civic assets), provides for the possibility of alienation to the occupiers of land already built on condition that the constructions are legitimately carried out or are condoned pursuant to the regulations must be in force regarding the amnesty of building violations.	The requirement that constructions be “legitimately carried out or condoned” under amnesty rules is dropped.
MISTRAL-7B	Subject Matter	the contested provisions violate the exclusive state competence in matters of environmental protection, since they authorize and regulate the construction of hotel and tourist structures in the state maritime territory, which is a limited resource and subject to particular environmental protection needs.	in allowing, at the same time, the construction of works that can lead to the consolidation of situations such as to hinder the complete development of landscape planning as well as the consumption of the coastal strips, landscape-bound, with fragmented interventions - violates the principle of the necessary planning of the assets subjected to landscape restrictions	Restricts any coastal works with tourism development, ignoring other types of works and the broader landscape planning context at issue in the target.
MISTRAL-7B	Terminology Precision	The contested provision provides for a “one-off” contribution for ASP workers without collective bargaining, in violation of the exclusive state competence in matters of civil law and the fundamental principles of coordination of public finance.	The regional law recognizes an economic emolument to ASP employees without agreement or consultation, violating the exclusive competence in civil law matters and replacing collective bargaining.	Alters the perceived nature of the benefit-“emolument” suggests ongoing pay-related entitlement, whereas “one-off contribution” reads as a discretionary grant.

LLM-as-a-Judge Factuality Evaluation. Given the high number of combinations of models and languages, we opt for an LLM-based evaluation protocol to evaluate the factual consistency of the generated explanation with respect to the target one. In particular, we drew on the methodological frameworks of G-Eval (Liu et al., 2023), Prometheus (Kim et al., 2024), and recent analyses of LLM-as-a-Judge reliability (Huang et al., 2024a), which collectively show that LLM evaluators, when properly constrained, correlate strongly with expert assessments on factuality and faithfulness tasks. All factuality scores for the conflict explanation task were produced using GPT-5-mini (OpenAI, 2025) under human supervision. Two authors manually inspected random subsets of the generated scores to verify alignment with human expectations and to correct cases where the evaluator’s reasoning was inconsistent with the articles’ legal meaning. The evaluator prompt was designed following the structured rubric paradigm of G-Eval, assigning an integer factuality score from 0 to 5 based solely on factual and legal correctness. Table 12 showcases the prompt used to evaluate all models.

Each evaluation was repeated across three independent passes, with scores averaged to reduce stochastic variation as recommended by Huang et al. (2024a). This setup allowed us to obtain reproducible, fine-grained factuality judgments at scale while maintaining the interpretability and reliability standards advocated in recent LLM-as-a-Judge research. The results in Fig. 9 illustrate how this factuality-oriented analysis reshapes the interpretation of explanation quality. Notably, closed-source models achieve markedly higher factuality scores, suggesting that their lower syntactic and semantic overlap with reference explanations primarily reflects stylistic differences rather than deficiencies in content preservation. Conversely, open-source models, even after fine-tuning, exhibit substantially lower factuality scores and frequent instances of hallucinated facts or legal inaccuracies.

Reference Retrieval. The results presented in Table 10 show a clear distinction between the performance of traditional models like BM25OKAPI and neural models across both Italian and English. BM25OKAPI

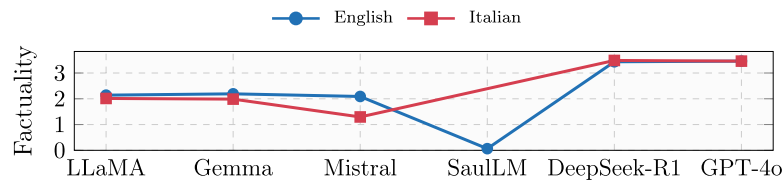


Fig. 9. Factualty score comparison across models for English and Italian splits.

demonstrates robust performance in Italian, likely due to its reliance on term matching rather than the linguistic nuances captured by neural models, which are often trained on English-dominated datasets. Neural models such as DISTILUSE, MINILM, and MPNET exhibit more competitive performance in English, where their training on rich and large-scale English corpora allows them to leverage semantic understanding effectively. This highlights the trade-off between traditional models' language-agnostic approach and neural models' reliance on high-quality training data in specific languages. However, when integrated into the HYBRID solution, they achieve the highest performance across both language subsets and all evaluation metrics.

5. Conclusion

This paper introduced CLASH-OF-LEGES, a novel bilingual dataset derived from real-world constitutional court rulings in Italy, specifically designed to address the challenges of legal conflict detection and explanation. We defined three core tasks—Conflict Detection, Conflict Explanation Generation, and Reference Retrieval—emphasizing their potential to advance AI applications in legal drafting, statutory analysis, and judicial review. The dataset bridges significant gaps in prior research by focusing on authentic conflicts between provisions in different legal documents rather than synthetically generated ones and incorporating bilingual support to broaden accessibility and research utility. Our experimental results highlight the performance of state-of-the-art multilingual and domain-specific LLMs across the proposed tasks. While these models achieved promising results, the findings also revealed persistent challenges, particularly in handling the nuanced language of legal texts and achieving robust performance in less-resourced languages like Italian. Our results underscore the unique challenges posed by our CLASH-OF-LEGES benchmark and the new capabilities that can be learned from it. Open-source models, having been exposed to the task through targeted fine-tuning, captured the nuanced linguistic and structural cues that differentiate conflictual from non-conflictual articles. In contrast, the closed-source models, evaluated without any in-task examples, consistently underperformed in conflict detection and explanation, indicating that their pre-training alone was insufficient to internalize these specialized patterns. This disparity persists despite the closed-source models' greater size and larger pre-training corpora, illustrating that sheer scale cannot substitute for task-specific adaptation. However, when explanations were assessed through an LLM factuality evaluation framework, a more nuanced picture was revealed. Closed-source models exhibited higher factual consistency and fewer hallucinated statements. This suggests that their lower lexical overlap and less elaborate argumentative structure stem more from stylistic divergence than from deficiencies in factual grounding. Taken together, these results show that scale and fine-tuning play complementary roles. Closed-source models provide stronger factual grounding, whereas fine-tuned open-source models excel at legal reasoning and interpretive explanation. The development and evaluation of CLASH-OF-LEGES mark a significant step toward automating the identification and resolution of legal conflicts. However, future work must focus on enhancing the reasoning capabilities and interpretability of AI models (Moro et al., 2024)—also with the aid of knowledge graph learning techniques (Frisoni et al., 2020)—and expanding datasets to include diverse jurisdictions and legal systems

to test cross-domain performance (Domeniconi et al., 2015; Lodi et al., 2010; Moro & Monti, 2012).

CRediT authorship contribution statement

Paolo Italiani: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing; **Gianluca Moro:** Conceptualization, Methodology, Investigation, Resources, Writing – review & editing, Supervision, Project administration; **Luca Ragazzi:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Research partially supported by: **AI-PACT** (CUP B47H22004450008, B47H22004460001); National Plan PNC-I.1 **DARE** initiative (PNC0000002, CUP B53C22006450001); PNRR Extended Partnership **FAIR** (PE00000013, Spoke 8); 2024 Scientific Research and High Technology Program, project “AI analysis for risk assessment of empty lymph nodes in endometrial cancer surgery”, the Fondazione Cassa di Risparmio in Bologna; Chips JU **TRISTAN** (G.A. 101095947). We thank Maggioli Group for co-funding the Ph.D. scholarship to P. Italiani.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.eswa.2025.130182](https://doi.org/10.1016/j.eswa.2025.130182).

References

- Abualhaija, S., Ceci, M., Sannier, N., Bianculli, D., Briand, L. C., Zetzsche, D., & Bodellini, M. (2024). Ai-enabled regulatory change analysis of legal requirements. In *2024 IEEE 32nd international requirements engineering conference (RE)* (pp. 5–17). IEEE.
- Ainslie, J., Lee-Thorp, J., de Jong, M., Zemlyanskiy, Y., Lebrón, F., & Sanghai, S. (2023). Gqa: Training generalized multi-query transformer models from multi-head checkpoints. *arXiv preprint arXiv:2305.13245*.
- Aires, J. P., Granada, R., Monteiro, J., Barros, R. C., & Meneguzzi, F. (2019a). Classifying norm conflicts using learned semantic representations. *CoRR*. <http://arxiv.org/abs/1906.02121>.
- Aires, J. P., Granada, R., Monteiro, J., Barros, R. C., & Meneguzzi, F. (2019b). Classifying norm conflicts using learned semantic representations. *arXiv preprint arXiv:1906.02121*.
- Aires, J. P., Granada, R. L., & Meneguzzi, F. R. (2019c). Concon: A contract conflict identifier. In *Proceedings of the 18th international conference on autonomous agents and multiagent systems, 2019, canadá*.
- Aires, J. P., Granada, R. L., Monteiro, J., Barros, R. C., & Meneguzzi, F. R. (2019d). Classification of contractual conflicts via learning of semantic representations. In *Proceedings of the 18th international conference on autonomous agents and multiagent systems (AAMAS), 2019, canadá*.
- Aires, J. P., & Meneguzzi, F. (2021). Norm conflict identification using a convolutional neural network. In *Coordination, organizations, institutions, norms, and ethics for governance of multi-agent systems XIII: International workshops COIN 2017 and COINE 2020, sao paulo, brazil, may 8–9, 2017 and virtual event, may 9, 2020, revised selected papers* (pp. 3–19). Springer.

- Aires, J. P., & Meneguzzi, F. R. (2017). A deep learning approach for norm conflict identification. In *Proceedings of the 16th international conference on autonomous agents and multiagent systems (AAMAS), 2017, brasil*.
- Aires, J. P., Monteiro, J., Granada, R., & Meneguzzi, F. (2018). Norm conflict identification using vector space offsets. In *2018 International joint conference on neural networks (IJCNN)* (pp. 1–8). IEEE.
- Aires, J. P., Pinheiro, D., Lima, V. S. d., & Meneguzzi, F. (2017). Norm conflict identification in contracts. *Artificial Intelligence and Law*, 25(4), 397–428.
- Alétrás, N., Tsarapatsanis, D., Preotiu-Pietro, D., & Lampos, V. (2016). Predicting judicial decisions of the european court of human rights: A natural language processing perspective. *PeerJ Computer Science*, 2, e93.
- Alkathery, E. R. (2023). Google translate errors in legal texts: Machine translation quality assessment. *SocArXiv*. March, 15.
- Alshamsan, A. R., & Chaudhry, S. A. (2022). Detecting privacy policies violations using natural language inference (NLI). In *2022 IEEE Asia-Pacific conference on computer science and data engineering (CSDE)* (pp. 1–6). IEEE.
- Araszkiewicz, M., Francesconi, E., & Zurek, T. (2021). Identification of contradictions in regulation. In *Legal knowledge and information systems* (pp. 151–160). IOS Press.
- Araszkiewicz, M., Francesconi, E., & Zurek, T. (2023). Identification of legislative errors. In *Proceedings of the nineteenth international conference on artificial intelligence and law* (pp. 2–11).
- Ashley, K. D. (2017). Artificial intelligence and legal analytics: new tools for law practice in the digital age. Cambridge University Press.
- Beltagy, I., Peters, M. E., & Cohan, A. (2020). Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*.
- Bird, S., & Loper, E. (2004). NLTK: The natural language toolkit. In *ACL* (pp. 214–217). Barcelona, Spain: ACL. <https://aclanthology.org/P04-3031>.
- Chalkidis, I., Fergadiotis, M., Tsarapatsanis, D., Alétrás, N., Androutsopoulos, I., & Malakasiotis, P. (2021a). Paragraph-level rationale extraction through regularization: A case study on european court of human rights cases. In K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, & Y. Zhou (Eds.), *Proceedings of the 2021 conference of the north american chapter of the association for computational linguistics: Human language technologies, NAACL-HLT 2021, online, june 6–11, 2021* (pp. 226–241). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2021.NAACL-MAIN.22>
- Chalkidis, I., Jana, A., Hartung, D., Bommarito, M., Androutsopoulos, I., Katz, D. M., & Alétrás, N. (2021b). LexGLUE: A benchmark dataset for legal language understanding in english. *arXiv preprint arXiv:2110.00976*.
- Chittimalli, P. K., & Anand, K. (2016). Domain-independent method of detecting inconsistencies in sbvr-based business rules. In *Proceedings of the international workshop on formal methods for analysis of business systems* (pp. 9–16).
- Chittimalli, P. K., Anand, K., Pradhan, S., Mitra, S., Prakash, C., Shere, R., & Naik, R. (2019). BuRito: A framework to extract, specify, verify and analyze business rules. In *2019 34th IEEE/ACM international conference on automated software engineering (ASE)* (pp. 1190–1193). IEEE.
- Colombo, P., Pires, T. P., Boudiaf, M., Culver, D., Melo, R., Corro, C., Martins, A. F. T., Esposito, F., Raposo, V. L., Morgado, S. et al. (2024). Saullm-7b: A pioneering large language model for law. *arXiv preprint arXiv:2403.03883*.
- Daniel Han, M. H., & team, U. (2023). Unsloth. <http://github.com/unslothai/unsloth>.
- DeepSeek-AI (2025). Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning. <https://arxiv.org/abs/2501.12948>.
- Domeniconi, G., Moro, G., Pagliarini, A., & Pasolini, R. (2015). Markov chain based method for in-domain and cross-domain sentiment classification. In A. L. N. Fred, J. L. G. Dietz, D. Aveiro, K. Liu, & J. Filipe (Eds.), *KDIR 2015 - Proceedings of the international conference on knowledge discovery and information retrieval, part of the 7th international joint conference on knowledge discovery, knowledge engineering and knowledge management (IC3k 2015), volume 1, lisbon, portugal, november 12–14, 2015* (pp. 127–137). SciTePress. <https://doi.org/10.5220/0005636001270137>
- Feng, X., Feng, X., & Qin, B. (2022). Msamsum: Towards benchmarking multi-lingual dialogue summarization. In S. Feng, H. Wan, C. Yuan, & H. Yu (Eds.), *Dialdoc@ACL 2022* (pp. 1–12). ACL. <https://doi.org/10.18653/v1/2022.dialdoc-1.1>
- Frisoni, G., Moro, G., & Carbonaro, A. (2020). Learning interpretable and statistically significant knowledge from unlabeled corpora of social text messages: A novel methodology of descriptive text mining. In S. Hammoudi, C. Quix, & J. Bernardino (Eds.), *Proceedings of the 9th international conference on data science, technology and applications, DATA 2020, lieusaint, paris, france, july 7–9, 2020* (pp. 121–132). SciTePress. <https://doi.org/10.5220/0009892001210132>
- Galassi, A., Drazewski, K., Lippi, M., & Torroni, P. (2020). Cross-lingual annotation projection in legal texts. In *Proceedings of the 28th international conference on computational linguistics* (pp. 915–926).
- He, C., Tan, T., Xue, S., & Tan, Y. (2023). Explaining legal judgments: A multitask learning framework for enhancing factual consistency in rationale generation. *Journal of King Saud University Computer and Information Sciences*, 35(10), 101868. <https://doi.org/10.1016/j.jksuci.2023.101868>
- He, C., Tan, T., Xue, S., & Tan, Y. (2025). Simulating judicial trial logic: Dual residual cross-attention learning for predicting legal judgment in long documents. *Expert Systems with Applications*, 261, 125462. <https://doi.org/10.1016/j.eswa.2024.125462>.
- Hematallah, H. (2021). Knowledge Extraction and Analysis of Medical Text with Particular Emphasis on Medical Guidelines. Ph.D. thesis. The University of North Carolina at Charlotte.
- Huang, H., Qu, Y., Liu, J., Yang, M., & Zhao, T. (2024a). An empirical study of llm-as-a-judge for LLM evaluation: Fine-tuned judge models are task-specific classifiers. *CoRR*. <https://doi.org/10.48550/ARXIV.2403.02839>
- Huang, S., Sun, J., & Li, R. (2024b). Neuralconflict: Using neural networks to identify norm conflicts in normative documents. *Expert Systems*, 41(6), e13035.
- Hurst, A., Lerer, A., Goucher, A. P. et al. (2024). Gpt-4o system card. *CoRR*. <https://doi.org/10.48550/ARXIV.2410.21276>
- Ichida, A. Y., & Meneguzzi, F. (2021). Detecting logical relation in contract clauses. *arXiv preprint arXiv:2111.01856*.
- Italiani, P., Moro, G., & Ragazzi, L. (2025). Enhancing legal question answering with data generation and knowledge distillation from large language models. *Artificial Intelligence and Law*. <https://doi.org/10.1007/s10506-025-09463-9>
- Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., de las, C. D., Bressand, F., Lengyel, G., Lample, G., Saulnier, L. et al. (2023). Mistral 7b. *arXiv preprint arXiv:2310.06825*.
- Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with gpus. *IEEE Transactions on Big Data*, 7(3), 535–547.
- Khattab, O., & Zaharia, M. (2020). Colbert: Efficient and effective passage search via contextualized late interaction over bert. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval* (pp. 39–48).
- Kim, S., Shin, J., Choi, Y., Jang, J., Longpre, S., Lee, H., Yun, S., Shin, S., Kim, S., Thorne, J., & Seo, M. (2024). Prometheus: Inducing fine-grained evaluation capability in language models. In *The twelfth international conference on learning representations, ICLR 2024, vienna, austria, may 7–11, 2024*. OpenReview.net. <https://openreview.net/forum?id=8euJaTveKw>.
- Koreeda, Y., & Manning, C. D. (2021). ContractNLI: A dataset for document-level natural language inference for contracts. *arXiv preprint arXiv:2110.01799*.
- van Leijenhorst, L., de Vries, A. P., Habben Jansen, T., & Wertheim, H. (2023). Sopalign: A tool for automatic estimation of compliance with medical guidelines. In *European conference on information retrieval* (pp. 307–312). Springer.
- Lin, C.-Y. (2004). ROUGE: A package for automatic evaluation of summaries. In *Text summarization branches out* (pp. 74–81). Barcelona, Spain: ACL. <https://aclanthology.org/W04-1013>.
- Liu, Y., Iter, D., Xu, Y., Wang, S., Xu, R., & Zhu, C. (2023). G-Eval: NLG evaluation using gpt-4 with better human alignment. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing, EMNLP 2023, singapore, december 6–10, 2023* (pp. 2511–2522). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.EMNLP-MAIN.153>
- Lodi, S., Moro, G., & Sartori, C. (2010). Distributed data clustering in multi-dimensional peer-to-peer networks. In H. T. Shen, & A. Bouguettaya (Eds.), *Database technologies 2010, twenty-first australasian database conference (ADC 2010), brisbane, australia, 18–22 january, 2010, proceedings* (pp. 171–178). Australian Computer Society (vol. 104). CRPIT. <http://portal.acm.org/citation.cfm?id=1862264&CFID=17470975&CFTOKEN=71845406>.
- Louis, A., Saxena, V., van Dijk, G., & Spanakis, G. O. Colbert-XM: A modular multi-vector representation model for zero-shot multilingual information retrieval. *CoRR* abs/2402.15059 (2024). <https://arxiv.org/abs/2402.15059>.
- Ma, Y., Shao, Y., Wu, Y., Liu, Y., Zhang, R., Zhang, M., & Ma, S. (2021). LecaRD: A legal case retrieval dataset for chinese law system. In F. Diaz, C. Shah, T. Suel, P. Castells, R. Jones, & T. Sakai (Eds.), *SIGIR '21: The 44th international ACM SIGIR conference on research and development in information retrieval, virtual event, canada, july 11–15, 2021* (pp. 2342–2348). ACM. <https://doi.org/10.1145/3404835.3463250>
- Mitra, S., Anand, K., & Chittimalli, P. K. (2018). Identifying anomalies in SBVR-based business rules using directed graphs and SMT-LBV2. In *Iceis (2)* (pp. 215–222).
- Moro, G., & Monti, G. (2012). W-Grid: A scalable and efficient self-organizing infrastructure for multi-dimensional data management, querying and routing in wireless data-centric sensor networks. *Journal of Network and Computer Applications*, 35(4), 1218–1234. <https://doi.org/10.1016/J.JNCA.2011.05.002>
- Moro, G., Piscaglia, N., Ragazzi, L., & Italiani, P. (2023a). Multi-language transfer learning for low-resource legal case summarization. *Artificial Intelligence and Law*, (pp. 1–29). <https://doi.org/10.1007/s10506-023-09373-8>
- Moro, G., & Ragazzi, L. (2022). Semantic self-segmentation for abstractive summarization of long documents in low-resource regimes. In *Thirty-sixth AAAI conference on artificial intelligence, AAAI 2022, thirty-fourth conference on innovative applications of artificial intelligence, IAAI 2022, the twelfth symposium on educational advances in artificial intelligence, EAAI 2022 virtual event, february 22, - march 1, 2022* (pp. 11085–11093). AAAI Press. <https://doi.org/10.1609/AAAI.V36I10.21357>
- Moro, G., & Ragazzi, L. (2023). Align-then-abstract representation learning for low-resource summarization. *Neurocomputing*, 548, 126356. <https://doi.org/10.1016/J.NEUCOM.2023.126356>
- Moro, G., Ragazzi, L., & Valgimigli, L. (2023b). Carburacy: Summarization models tuning and comparison in eco-sustainable regimes with a novel carbon-aware accuracy. In B. Williams, Y. Chen, & J. Neville (Eds.), *Thirty-seventh AAAI conference on artificial intelligence, AAAI 2023, thirty-fifth conference on innovative applications of artificial intelligence, IAAI 2023, thirteenth symposium on educational advances in artificial intelligence, EAAI 2023, washington, dc, usa, february 7–14, 2023* (pp. 14417–14425). AAAI Press. <https://doi.org/10.1609/AAAI.V37I12.26686>
- Moro, G., Ragazzi, L., & Valgimigli, L. (2023c). Graph-based abstractive summarization of extracted essential knowledge for low-resource scenarios. In K. Gal, A. Nowé, G. J. Nalepa, R. Fairstein, & R. Radulescu (Eds.), *ECAI 2023 - 26th European conference on artificial intelligence, september 30, - october 4, 2023, kraków, poland - including 12th conference on prestigious applications of intelligent systems (PAIS 2023)* (pp. 1747–1754). IOS Press (vol. 372). Frontiers in Artificial Intelligence and Applications. <https://doi.org/10.3233/FAIA230460>
- Moro, G., Ragazzi, L., Valgimigli, L., Vincenzi, F., & Freddi, D. (2024). Revelio: Interpretable long-form question answering. In *The second tiny papers track at ICLR 2024, tiny papers @ ICLR 2024, vienna, austria, may 11, 2024*. OpenReview.net. <https://openreview.net/forum?id=fyvEJXsaQf>.
- Muennighoff, N., Wang, T., Sutawika, L., Roberts, A., Biderman, S., Scao, T. L., Bari, M. S., Shen, S., Yong, Z. X., Schölkopf, H., Tang, X., Radev, D., Aji, A. F., Almubarak, K., Albanie, S., Alyafeai, Z., Webson, A., Raff, E., & Raffel, C. (2022). Crosslingual gener-

- alization through multitask finetuning. *CoRR*. <https://doi.org/10.48550/ARXIV.2211.01786>
- OpenAI (2025). Gpt-5 system card. <https://cdn.openai.com/gpt-5-system-card.pdf>.
- Rabelo, J., Kim, M.-Y., Goebel, R., Yoshioka, M., Kano, Y., & Satoh, K. (2021). Coliee 2020: Methods for legal document retrieval and entailment. In *New frontiers in artificial intelligence: JSAI-IAI 2020 workshops, JURISIN, LENLS 2020 workshops, virtual event, november 15–17, 2020, revised selected papers 12* (pp. 196–210). Springer.
- Ragazzi, L., Moro, G., Guidi, S., & Frisoni, G. (2024). Lawsuit: A large expert-written summarization dataset of italian constitutional court verdicts. *Artificial Intelligence and Law*. <https://api.semanticscholar.org/CorpusID:272530787>.
- Ragazzi, L., Moro, G., Valgimigli, L., & Fiorani, R. (2025). Cross-document distillation via graph-based summarization of extracted essential knowledge. *IEEE Transactions on Audio, Speech and Language Processing*, 33, 518–527. <https://doi.org/10.1109/TASLP.2024.3490375>
- Reimers, N., & Gurevych, I. (2019a). Sentence-bert: Sentence embeddings using siamese bert-networks. *arXiv preprint arXiv:1908.10084*.
- Reimers, N., & Gurevych, I. (2019b). Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 conference on empirical methods in natural language processing*. Association for Computational Linguistics. <http://arxiv.org/abs/1908.10084>.
- Rissland, E. L., & Ashley, K. D. (1987). A case-based system for trade secrets law. In *Proceedings of the 1st international conference on artificial intelligence and law* (pp. 60–66).
- Robertson, S. E., Walker, S., Jones, S., Hancock-Beaulieu, M. M., Gatford, M. et al. (1995). Okapi at TREC-3. *Nist Special Publication Sp*, 109, 109.
- Sai, C., Winter, K., Fernanda, E., & Rinderle-Ma, S. (2023). Detecting deviations between external and internal regulatory requirements for improved process compliance assessment. In *International conference on advanced information systems engineering* (pp. 401–416). Springer.
- Scao, T. L., Fan, A., Akiki, C., Pavlick, E., Ilıc, S., Hesslow, D., Castagné, R., Luccioni, A. S., Yvon, F., Gallé, M., Tow, J., Rush, A. M., Biderman, S., Webson, A., Ammanamanchi, P. S., Wang, T., Sagot, B., Muennighoff, N., del Moral, A. V., Ruwase, O., Bawden, R., Bekman, S., McMillan-Major, A., Beltagy, I., Nguyen, H., Saulnier, L., Tan, S., Suarez, P. O., Sanh, V., Laurençon, H., Jernite, Y., Launay, J., Mitchell, M., Raffel, C., Gokaslan, A., Simhi, A., Soroa, A., Aji, A. F., Alfassy, A., Rogers, A., Nitzav, A. K., Xu, C., Mou, C., Emezue, C., Klammer, C., Leong, C., van Strien, D., & Adelani, D. I., et al. (2022). BLOOM: A 176b-parameter open-access multilingual language model. *CoRR*. <https://doi.org/10.48550/ARXIV.2211.05100>
- Schumann, G., & Gómex, J. M. (2024). Detection of conflicts, contradictions and inconsistencies in regulatory documents: A literature review. In *2024 Fifth international conference on intelligent data science technologies and applications (IDSTA)* (pp. 81–88). IEEE.
- Song, K., Tan, X., Qin, T., Lu, J., & Liu, T. (2020). Mpnnet: Masked and permuted pre-training for language understanding. In H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, & H. Lin (Eds.), *Advances in neural information processing systems 33: Annual conference on neural information processing systems 2020, neurIPS 2020, december 6–12, 2020, virtual*.
- Team, G., Riviere, M., Pathak, S., Sessa, P. G., Hardin, C., Bhupatiraju, S., Hussenot, L., Mesnard, T., Shahriari, B., Ramé, A. et al. (2024). Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.
- Tiersma, P. (2000). Legal language. *Bibliovault OAI Repository, the University of Chicago Press*, 27. [https://doi.org/10.1016/S1352-0237\(00\)00210-0](https://doi.org/10.1016/S1352-0237(00)00210-0)
- Tugener, D., von Däniken, P., Peetz, T., & Cieliebak, M. (2020). LEDGAR: A large-scale multi-label corpus for text classification of legal provisions in contracts. In N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds.), *Proceedings of the 12th language resources and evaluation conference, LREC 2020, marseille, france, may 11–16, 2020* (pp. 1235–1241). European Language Resources Association. <https://aclanthology.org/2020.lrec-1.155/>.
- Ubayawardena, N., & Hansani, M. (2023). A study on the effectiveness of using google translate in legal translation: With special reference to selected legal documents of the registrar general's department, 1.
- Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., & Zhou, M. (2020). Minilm: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. *Advances in Neural Information Processing Systems*, 33, 5776–5788.
- WIESMANN, E. (2019). Machine translation in the field of law: A study of the translation of italian legal texts into german. *Comparative Legilinguistics*, 37, 117–153. <https://pressto.amu.edu.pl/index.php/cl/article/view/18273>. <https://doi.org/10.14746/cl.2019.37.4>
- Yuan, W., Neubig, G., & Liu, P. (2021). Bartscore: Evaluating generated text as text generation. In M. Ranzato, A. Beygelzimer, Y. N. Dauphin, P. Liang, & J. W. Vaughan (Eds.), *Advances in neural information processing systems 34: annual conference on neural information processing systems 2021, neurIPS 2021, december 6–14, 2021, virtual* (pp. 27263–27277). <https://proceedings.neurips.cc/paper/2021/hash/e4d2b6e6fdeca3e60e0f1a62fee3d9dd-Abstract.html>.
- Zadrozny, W., Hematiam, H., & Garbayo, L. (2017). Towards semantic modeling of contradictions and disagreements: A case study of medical guidelines. In C. Gardent, & C. Retoré (Eds.), *IWCS 2017 - 12th international conference on computational semantics - short papers, montpellier, france, september 19. - 22, 2017*. The Association for Computer Linguistics. <https://aclanthology.org/W17-6943/>.
- Zheng, L., Guha, N., Anderson, B. R., Henderson, P., & Ho, D. E. (2021). When does pre-training help?: Assessing self-supervised learning for law and the caseHOLD dataset of 53, 000+ legal holdings. In J. Maranhão, & A. Z. Wyner (Eds.), *ICAIL '21: Eighteenth international conference for artificial intelligence and law, são paulo brazil, june 21. - 25, 2021* (pp. 159–168). ACM. <https://doi.org/10.1145/3462757.3466088>
- Zhong, H., Xiao, C., Tu, C., Zhang, T., Liu, Z., & Sun, M. (2020). JEC-QA: A legal-domain question answering dataset. In *The thirty-fourth AAAI conference on artificial intelligence, AAAI 2020, the thirty-second innovative applications of artificial intelligence conference, IAAI 2020, the tenth AAAI symposium on educational advances in artificial intelligence, AAAI 2020, new york, ny, usa, february 7–12, 2020* (pp. 9701–9708). AAAI Press. <https://doi.org/10.1609/AAAI.V34I05.6519>
- Zhu, J., Wang, Q., Wang, Y., Zhou, Y. et al. (2019). NCLS: Neural cross-lingual summarization. In *EMNLP-IJCNLP* (pp. 3054–3064). Hong Kong, China: ACL. <https://doi.org/10.18653/v1/D19-1302>